

Stars and Stellar Evolution Lab Assignment

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1 Model Resolution and Time Scales

1.1 Why is there only one history file but multiple profile files?

The MESA output consists of a single history file and multiple profile files because they serve fundamentally different structural and diagnostic purposes. The history file records global stellar properties as a function of time throughout the entire evolution. Each row corresponds to an individual, sequentially generated stellar model number and contains time-dependent global quantities such as stellar age, total luminosity, effective temperature, total radius, central temperature, central density, and central core chemical abundances. Consequently, the history file provides a continuous macroscopic record of the evolutionary track of the star.

In contrast, profile files reveal the one-dimensional internal structure of the star at highly localized, specific evolutionary stages. Each profile stores spatial quantities as a function of the internal mesh zone index or mass coordinate m , including local density, temperature, pressure, interior chemical composition fractions, and active convective or radiative mixing properties throughout the stellar interior. Because the internal structural profiles change drastically during evolution, multiple profile files are saved only at selected model snapshots.

Therefore, only one history file is required to track the long-term temporal evolution of global stellar parameters, whereas multiple profile files are needed to capture structural snapshots of the stellar interior at critical evolutionary epochs.

1.2 Which of the phases mentioned above did the simulation go through? Graphically illustrate your answer.

The evolutionary phases of the stellar model were identified using both the Hertzsprung-Russell (HR) diagram and the internal nuclear quantities stored in the MESA history file. The HR diagram illustrates the global track of the star, while the central hydrogen abundance and nuclear luminosities reveal the underlying microphysical processes driving the observed macro-evolutionary features.

The model begins on the Pre-Main Sequence (PMS), during which gravitational contraction provides the dominant energy source. At this stage, the central hydrogen abundance remains essentially constant because the core temperature has not yet reached the threshold necessary to ignite nuclear fusion. The star contracts under its own gravity, its effective temperature steadily shifts, and its luminosity drops as it approaches the main sequence.

The model reaches the Zero-Age Main Sequence (ZAMS) when core nuclear energy generation becomes sufficient to completely support the stellar luminosity against gravitational collapse, matching approximately:

$$\frac{L_{\text{nuc}}}{L} > 0.99 \quad (1)$$

From this point onward, the star enters the core hydrogen-burning phase (the Main Sequence), during which hydrogen is steadily converted into helium in the core primarily through the proton-proton chain. This phase occupies the vast majority of the stellar lifetime and is characterized by a gradual, monotonic decrease in the central hydrogen mass fraction from approximately 0.70 down to nearly zero.

After the complete exhaustion of central hydrogen, the model leaves the main sequence. The inert helium core contracts under its own weight while hydrogen fusion continues in a shell surrounding the core, causing the stellar envelope to expand rapidly and the luminosity to climb. This marks the onset of hydrogen shell burning and the transition toward the subgiant and red giant branch.

We defined the Terminal-Age Main Sequence (TAMS) by the core condition:

$$X_c(\text{H}) \approx 0.001 \quad (2)$$

Evolution of Stellar Properties of $1M_{\odot}$ star

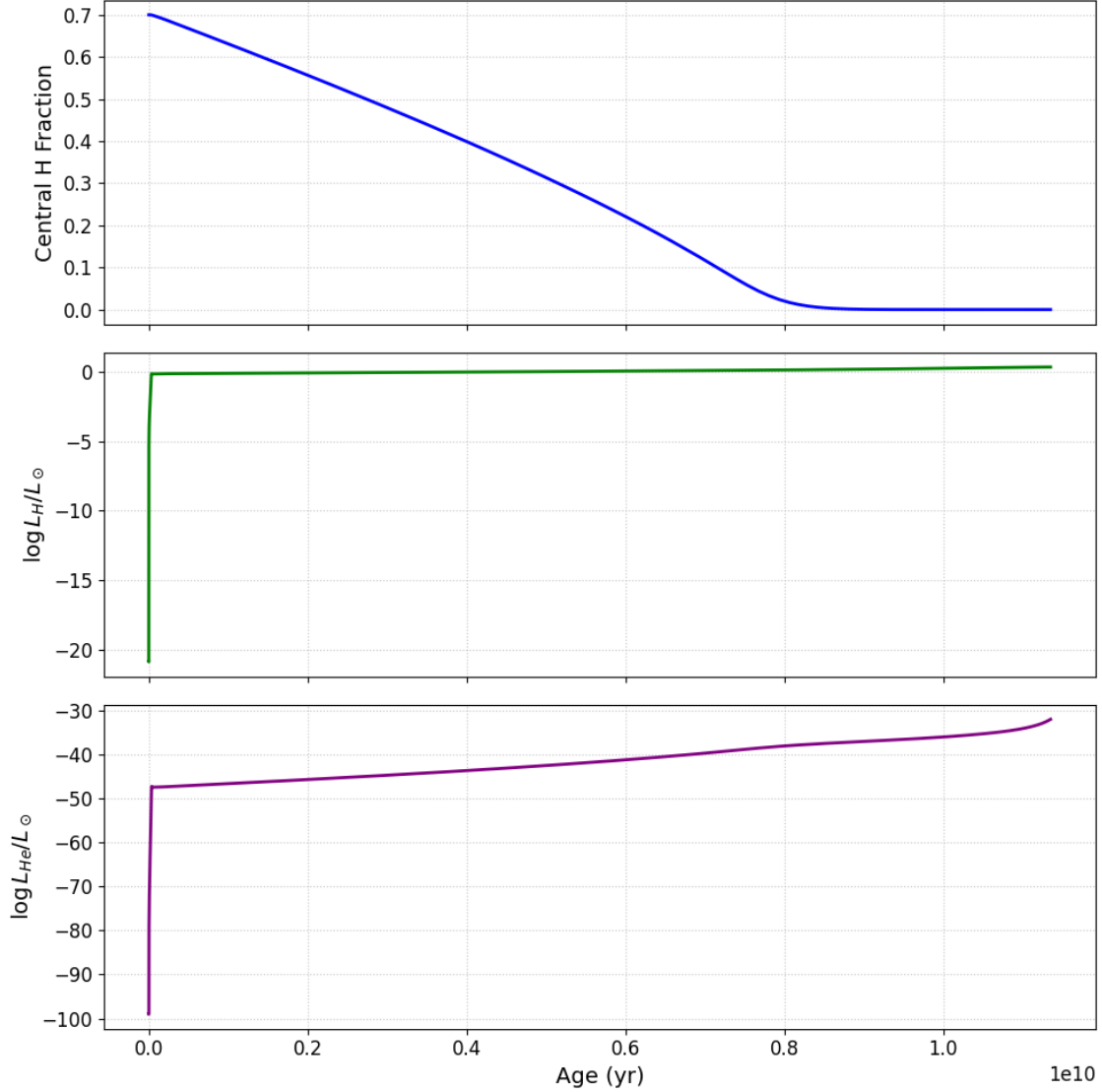


Figure 1: Evolutionary tracks of internal properties for the $1 M_{\odot}$ stellar model as a function of age. **Top panel:** The monotonic decay of the central hydrogen mass fraction from $X_c \approx 0.7$ down to zero represents steady main-sequence core burning. **Middle panel:** The hydrogen-burning luminosity ($\log L_H / L_{\odot}$) rapidly shoots up from near-zero values (≈ -21) during pre-main-sequence contraction to a stable plateau near $\log L_H / L_{\odot} \approx 0$, demonstrating that hydrogen fusion serves as the absolute dominant energy source following the PMS phase. **Bottom panel:** The helium-burning luminosity ($\log L_{He} / L_{\odot}$) remains completely negligible (≤ -30) throughout the run, confirming that the simulation terminated before reaching core helium ignition conditions.

At this point, hydrogen burning in the core effectively ceases. Following hydrogen exhaustion, the model enters the shell hydrogen-burning phase. The growth of the helium core and the increase in shell-burning luminosity cause the stellar radius to expand and the effective temperature to decrease. Consequently, the star moves toward the red-giant branch in the HR diagram. No core helium-burning phase was reached during this simulation because the run terminated shortly after core hydrogen exhaustion.

Therefore, the evolutionary phases traversed by the model were:

1. Pre-main-sequence contraction.
2. Core hydrogen burning (main sequence).
3. Hydrogen shell burning following core hydrogen exhaustion.

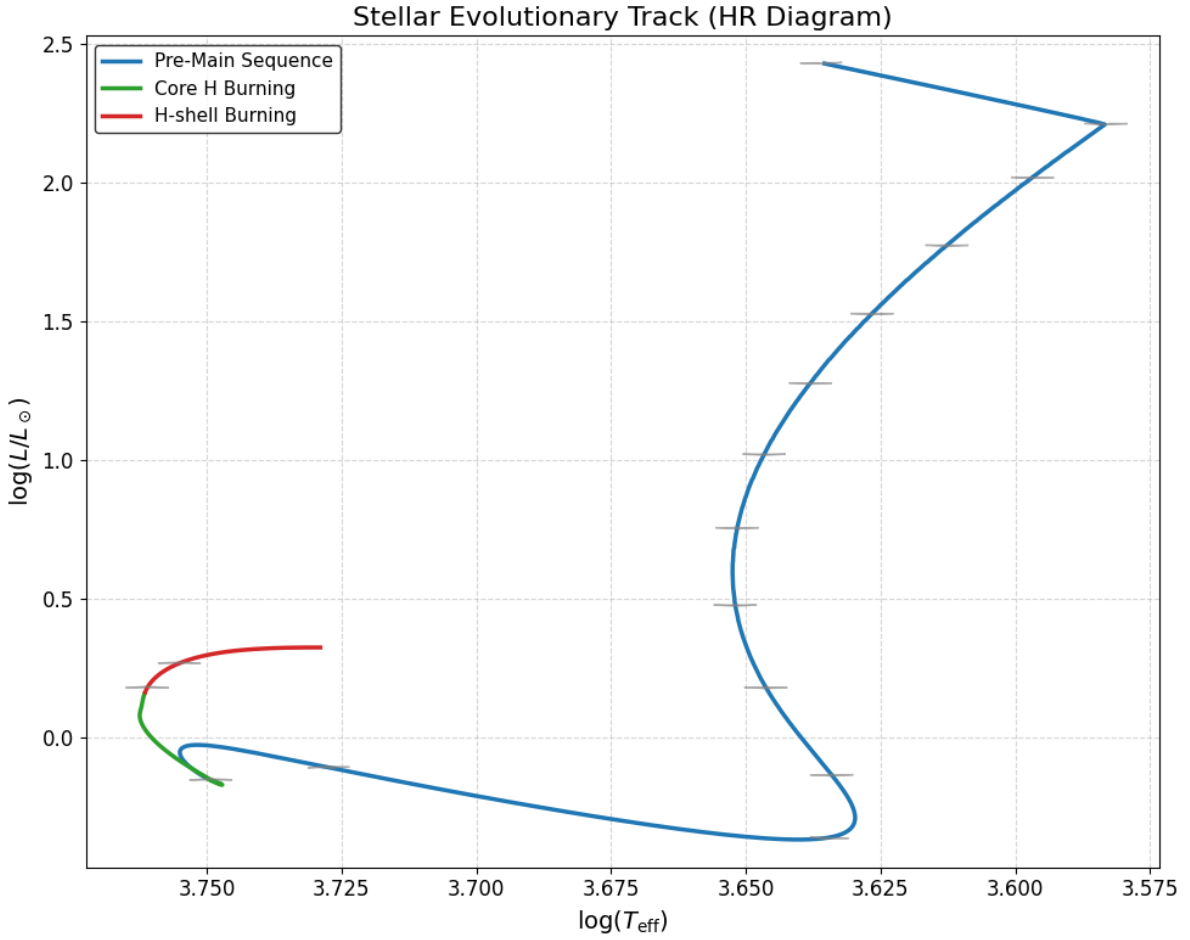


Figure 2: Hertzsprung-Russell (HR) diagram mapping the evolutionary track ($\log L/L_{\odot}$ vs. $\log T_{\text{eff}}$) of the $1 M_{\odot}$ stellar model. The color-coded segments visually distinguish the distinct physical regimes traversed during the simulation: the blue path traces the rapid luminosity drop during Pre-Main Sequence (PMS) gravitational contraction; the green segment marks the stable core hydrogen-burning Main Sequence phase stretching up to the core exhaustion point; and the red segment illustrates the onset of envelope expansion during the Hydrogen Shell burning phase as the star transitions toward the red giant branch.

The following table records some calculated evolutionary properties:

Table 1: Calculated Evolutionary Properties for a $1 M_{\odot}$ Stellar Model

Property	Calculated Value
ZAMS Age	3.86×10^8 yrs
TAMS Age	8.91×10^9 yrs
Shell Age	9.18×10^9 yrs
L_{nuc}/L at ZAMS	0.9994
Center H at ZAMS	0.6995
Center H at TAMS	8.6×10^{-4}
He core mass at shell	$0.0103 M_{\odot}$

1.3 Show with a plot whether or not the size of the time steps in the history file changes during the simulation. If it does: why?

The timestep used by MESA is not constant throughout the stellar evolution calculation but is adjusted automatically through an adaptive timestep algorithm. The timestep was obtained from the `log_dt` column of the history file using the logarithmic relation:

$$\Delta t = 10^{\log dt} \quad (3)$$

Figure 3 shows that the timestep varies by more than twelve orders of magnitude during the simulation. At the beginning of the calculation, during the Pre-main-sequence contraction phase, the timestep is kept extremely small because the stellar structure changes rapidly and the numerical solver requires high temporal resolution to reach mathematical convergence. As the star approaches and moves along the main sequence, structural changes slow down drastically, allowing MESA to increase the timestep size substantially to minimize total computational cost.

Near the end of the main sequence, after the exhaustion of central hydrogen, the stellar structure begins to evolve rapidly once more as the helium core contracts and hydrogen shell burning develops. Consequently, the timestep decreases to maintain numerical accuracy and stable convergence of the solution. The variation of the timesteps therefore reflects the adaptive timestep control implemented by MESA, which automatically balances computational efficiency with numerical precision.

1.4 Consider the grid spacing in a profile file of your choice in terms of numerical cells per unit mass. Show with a plot where the spacing is the most dense near the surface or near the core? Why is the grid spacing not constant?

MESA employs adaptive mesh calculations to manage computational resources efficiently. The internal stellar structure is highly complex and evolves non-uniformly; individual zones change at vastly different rates depending on local physical variables. Internal regions experiencing steep temperature gradients, major composition boundaries, local nuclear burning, or convective cell mixing require high spatial resolution, whereas uniform intermediate zones can be tracked using far fewer grid points. We calculated the local mass spacing δm between adjacent grid cells using the enclosed mass distribution data.

The mesh resolution is obtained by taking the inverse of the mass difference between consecutive shell edges, yielding:

$$\text{Mesh Resolution} = \frac{1}{\delta m} \quad (4)$$

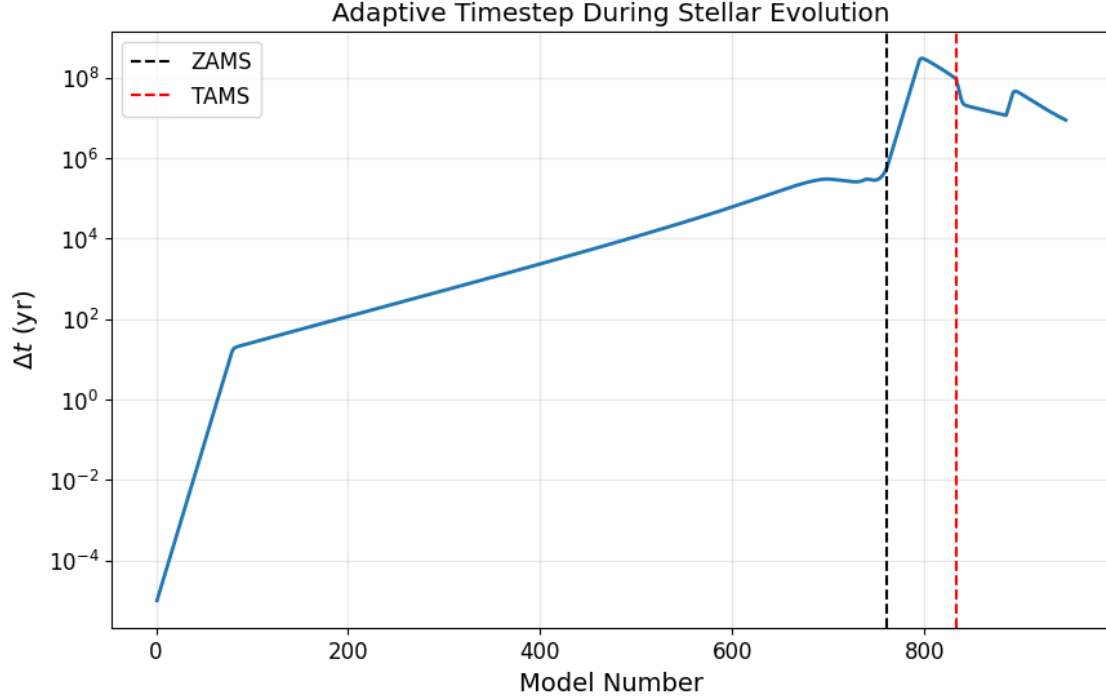


Figure 3: Evolution of the adaptive timestep size (Δt in years) plotted against model number (left panel) and stellar age in years (right panel). The adaptive control scheme automatically restricts Δt to extremely low fractions ($\approx 10^{-5}$ years) during the highly dynamic early Pre-Main Sequence phase to resolve rapid structural changes. As the star settles into a stable hydrostatic and thermal equilibrium on the Main Sequence, the timestep size smoothly scales up by several orders of magnitude ($\approx 10^5$ to 10^8 years) due to minimal structural activity. The distinct dips and features visible around the ZAMS and TAMS transitions demonstrate the algorithm tightening the temporal resolution to safely handle rapid physical re-adjustments and core fuel depletion.

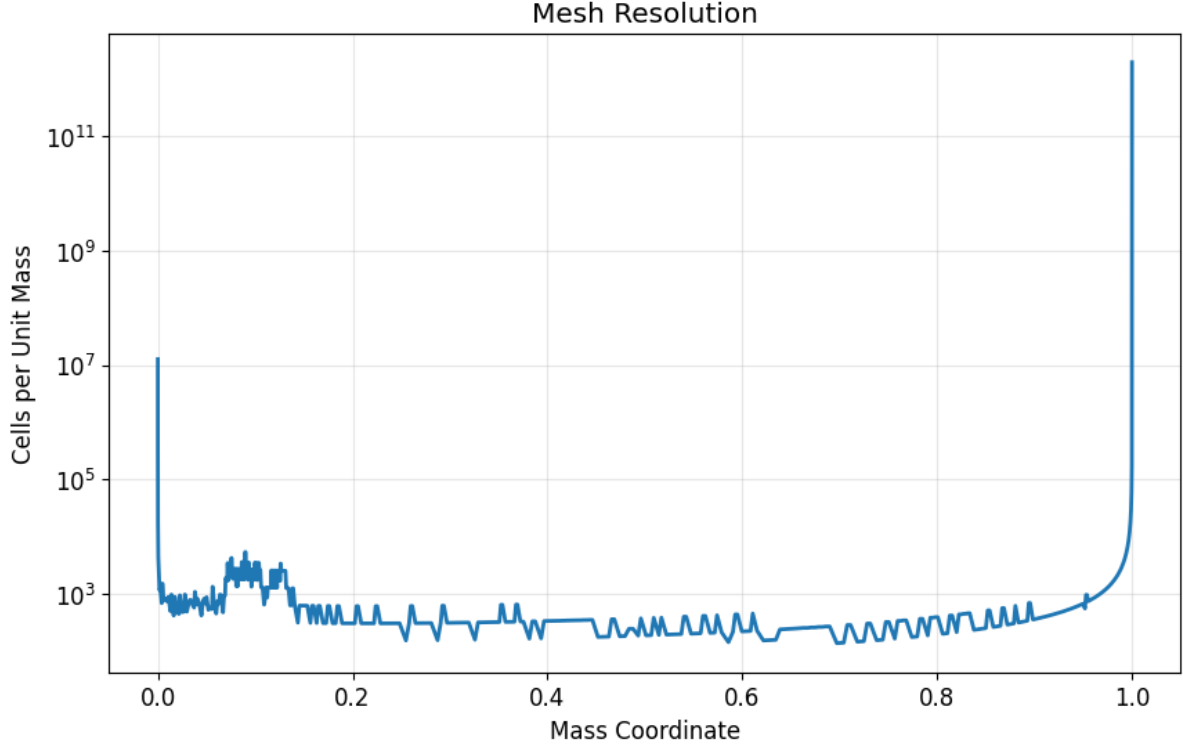


Figure 4: Stellar mesh density expressed in cells per unit mass as a function of the normalized interior mass coordinate m/M . Two highly refined structural peaks are clearly visible. The sharp peak at the core ($m/M \rightarrow 0$) yields a high cell concentration ($\sim 10^7$) to resolve intense nuclear energy generation and sharp chemical gradients. The prominent peak at the stellar boundary ($m/M \rightarrow 1$) reaches the highest refinement level ($\sim 10^{12}$ cells per unit mass) to capture steep atmospheric pressure profiles, opacity variations, and expansion features near the vacuum edge.

We observe two distinct spatial resolution peaks in Figure 4: one situated at the stellar core ($m/M \rightarrow 0$) and a second, much steeper peak close to the outermost surface layer ($m/M \rightarrow 1$). The surface layer requires the highest overall refinement. This is because surface boundary layers feature rapidly shifting gas densities, pressures, and opacities over tiny fractions of total mass, all while dictating the definitive macroscopic radius of the star.

Meanwhile, high mesh resolution at the core is necessary to capture sensitive temperature-dependent nuclear reactions and shifting chemical boundaries. The smooth, intermediate regions are physically quiet and require fewer cells, allowing MESA to maximize numerical accuracy while avoiding unnecessary computational expense.

1.5 During the evolution of the stellar model, on what timescales does the contraction/expansion take place?

A star is governed by competing mechanisms of gravity, pressure, energy generation, and energy transport, which operate on distinct, decoupled timescales.

The **Dynamical Timescale** (τ_{dyn}) represents the time required for a star to collapse or expand if its hydrostatic balance were completely disrupted:

$$\tau_{\text{dyn}} = \sqrt{\frac{R^3}{GM}} \quad (5)$$

The **Thermal (Kelvin–Helmholtz) Timescale** (τ_{KH}) characterizes the time a star takes to radiate away its entire reservoir of internal gravitational potential energy at its current luminosity output:

$$\tau_{\text{KH}} = \frac{GM^2}{RL} \quad (6)$$

The **Nuclear Timescale** (τ_{nuc}) defines the maximum lifetime supported by core nuclear fuel reserves. Assuming only hydrogen fusion converts roughly 0.7% of mass into energy, and that approximately 10% of the total stellar mass is available within the hot core, it is expressed as:

$$\tau_{\text{nuc}} = \frac{0.007 \times 0.1Mc^2}{L} = \frac{0.0007Mc^2}{L} \quad (7)$$

The **Radius Evolution Timescale** (τ_R) measures the actual characteristic rate of physical contraction or expansion of the stellar radius:

$$\tau_R = \left| \frac{R}{\dot{R}} \right| \quad (8)$$

The numerical results confirm the expected structural hierarchy $\tau_{\text{dyn}} \ll \tau_{\text{KH}} \ll \tau_{\text{nuc}}$ throughout the simulation. The dynamical timescale remains extremely short ($\sim 10^{-4}$ years), which indicates that any mechanical perturbation is restored almost instantaneously by the internal pressure gradient.

During the Pre-main-sequence contraction, the radius-change timescale closely matches the Kelvin–Helmholtz timescale, demonstrating that early stellar development is driven by gravitational energy release. After hydrogen ignition, the star settles onto the Main Sequence; the structural radius changes very slowly, and the evolution proceeds on the long nuclear timescale.

Near the end of the main sequence, when central hydrogen is exhausted and shell burning sets in, the internal structure undergoes rapid re-adjustment and the radius timescale drops sharply. This confirms that post-main-sequence evolution transitions away from nuclear control back toward rapid thermal and structural adjustments.

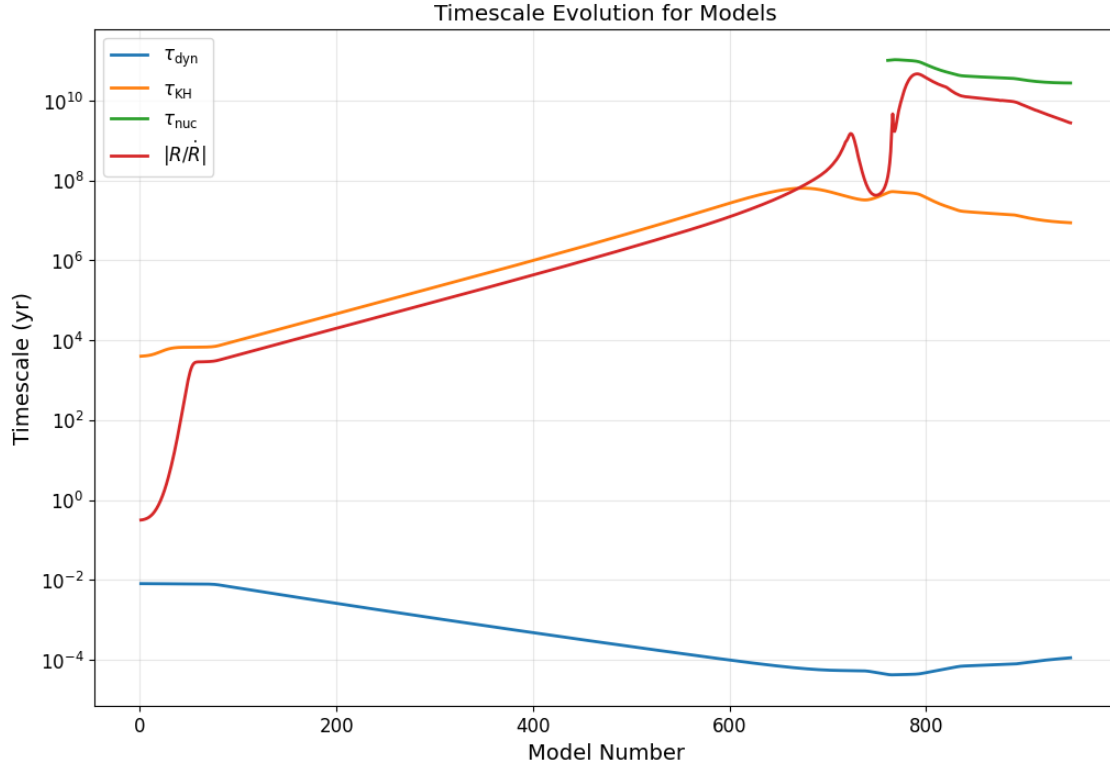


Figure 5: Evolutionary behavior of the four fundamental stellar timescales (τ_{dyn} , τ_{KH} , τ_{nuc} , and the structural radius timescale $|R/\dot{R}|$) plotted against the ongoing model sequence number. The strict mathematical hierarchy $\tau_{\text{dyn}} \ll \tau_{\text{KH}} \ll \tau_{\text{nuc}}$ is maintained across the entire life cycle. The dynamical timescale remains consistently small ($\sim 10^{-2}$ to 10^{-4} years), confirming that macroscopic hydrostatic equilibrium is securely maintained at every step of the evolution.

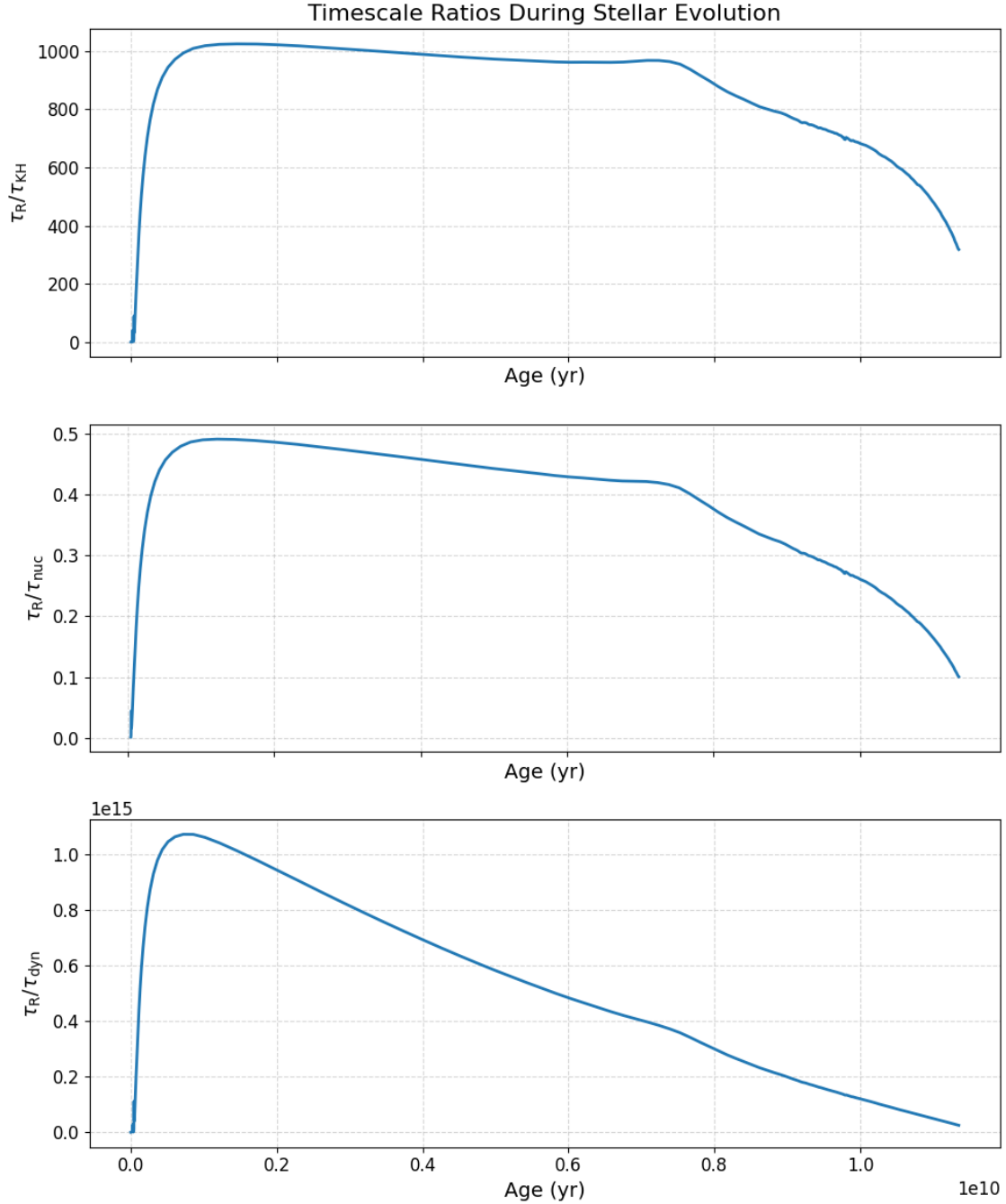


Figure 6: Stellar timescale ratios (τ_R/τ_{KH} , τ_R/τ_{nuc} , and τ_R/τ_{dyn}) tracked as a function of chronological stellar age in years. **Top panel:** During the initial pre-main-sequence stage, the radius timescale sits within a fixed order of magnitude of the thermal timescale ($\tau_R/\tau_{KH} \sim 10^2 - 10^3$), proving that gravitational contraction operates on a thermal adjustment path. **Middle panel:** On the Main Sequence, the ratio approaches unity relative to the nuclear timeline ($\tau_R/\tau_{nuc} \sim 0.1 - 0.5$), highlighting that core hydrogen consumption dictates the structural evolution. **Bottom panel:** The ratio τ_R/τ_{dyn} remains immensely large ($\sim 10^{15}$), showing that radius adjustments happen quasi-statically.

2 Convective Cores and Advanced Evolutionary Stages

2.1 Convective Regions at TAMS (Q2.1)

The internal convective boundaries of the massive $47.504 M_{\odot}$ stellar model at the Terminal-Age Main Sequence (TAMS) were determined precisely via the Schwarzschild stability criterion. In stellar interiors, a local zone undergoes convective mixing when the radiative temperature gradient (∇_{rad}) exceeds the adiabatic temperature gradient (∇_{ad}):

$$\nabla_{\text{rad}} > \nabla_{\text{ad}} \quad (9)$$

Figure 7 displays the spatial distribution of these temperature gradients evaluated across the internal profile at the TAMS epoch. The radiative gradient exhibits a steep, non-linear profile near the deep interior, falling smoothly toward the outer layers. The boundary separating the convective core from the radiative envelope is defined by the intersection point where $\nabla_{\text{rad}} = \nabla_{\text{ad}}$.

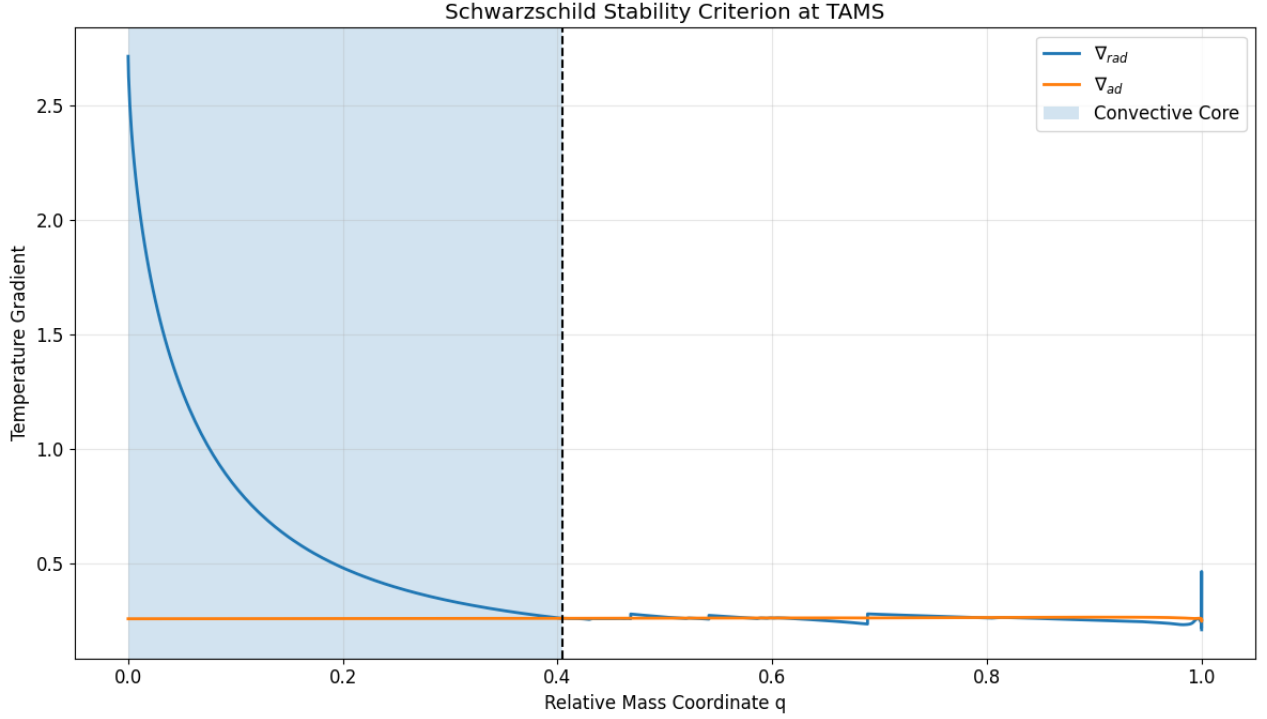


Figure 7: Internal temperature gradients (∇_{rad} in blue and ∇_{ad} in orange) plotted as a function of the relative interior mass coordinate $q = m/M$ for the $47.504 M_{\odot}$ stellar model at the terminal-age main sequence (TAMS). The light blue shaded region explicitly highlights the active convective core environment extending from the center up to the localized transition boundary at $q \approx 0.405$, where the Schwarzschild instability condition ($\nabla_{\text{rad}} > \nabla_{\text{ad}}$) is structurally satisfied.

This crossover condition occurs precisely at the relative mass coordinate:

$$q \approx 0.405 \quad (10)$$

Consequently, the convective core encompasses the innermost 40.5% of the total stellar mass, yielding a bounded region defined by $0 \leq q \leq 0.405$. This corresponds to an absolute convective core mass of:

$$M_{\text{cc}} \approx 19.24 M_{\odot} \quad (11)$$

The presence of a massive, highly extended convective core is driven by the extreme temperature dependence of the Carbon-Nitrogen-Oxygen (CNO) cycle, which serves as the absolute dominant energy

generation pathway in high-mass stars. Because the nuclear energy generation rate scales steeply with core temperature ($\epsilon_{\text{CNO}} \propto T^{16}$ to T^{20}), energy production is confined into a highly concentrated central volume. This localized, enormous energy release generates a large radiative flux that cannot be transported efficiently by radiation alone, causing ∇_{rad} to swell far beyond ∇_{ad} and forcing the development of a turbulent convective core.

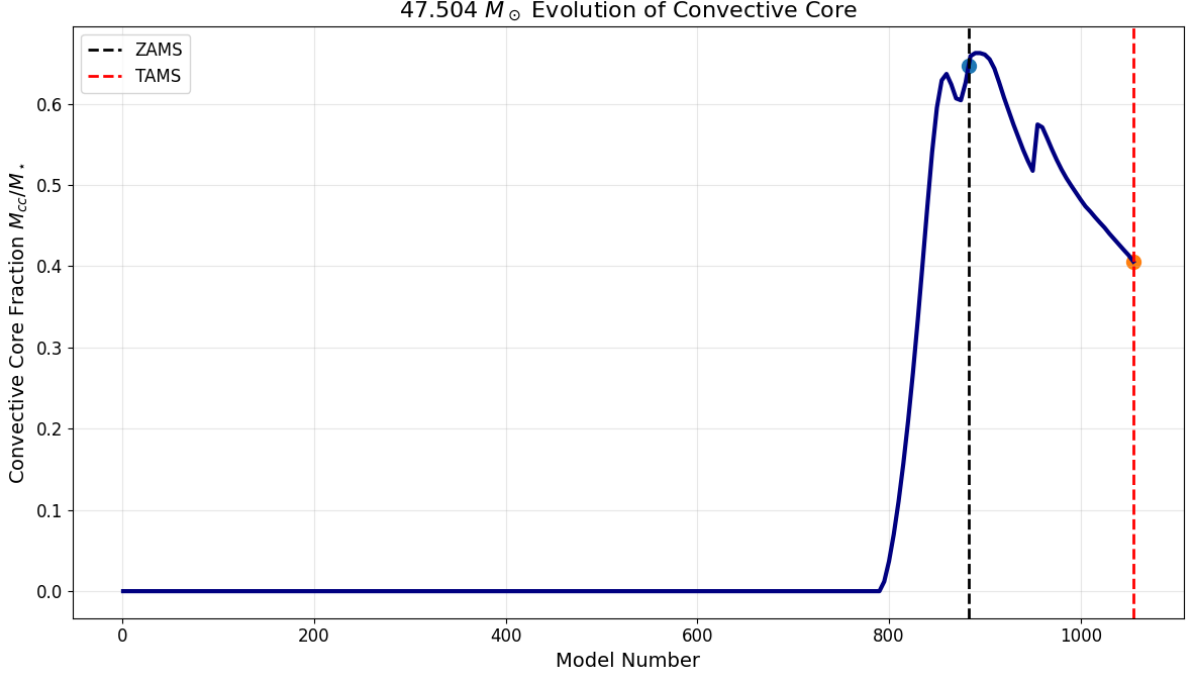


Figure 8: Temporal development of the convective core mass fraction (M_{cc}/M_*) plotted as a function of the sequentially generated model numbers for the $47.504 M_{\odot}$ stellar run. The core fraction remains at zero during early pre-main-sequence contraction before abruptly opening up and peaking at $M_{\text{cc}}/M_* \approx 0.65$ upon core hydrogen ignition at the Zero-Age Main Sequence (ZAMS). As hydrogen fuel is steadily consumed, the core boundaries shrink monotonically over time, contracting down to $M_{\text{cc}}/M_* \approx 0.405$ at the Terminal-Age Main Sequence (TAMS) marker.

2.2 Convective Core Boundaries Across the High-Mass Spectrum (Q2.2)

To evaluate the collective behavior of convective zones at the Terminal-Age Main Sequence (TAMS), the compiled class dataset was synthesized into a comparative structural scatter plot in Figure 9. The cohort focuses strictly on the high-mass stellar regime, capturing initial masses from approximately $10 M_{\odot}$ up to nearly $50 M_{\odot}$. Because every model in this high-mass range is powered by core nuclear fusion up to the point of central hydrogen exhaustion, each star exhibits a central convective core extending directly from the absolute center ($q_{\text{bottom}} = 0$) out to an upper boundary mass coordinate ($q = q_{\text{top}}$). Consequently, the graph directly tracks the maximum mass fraction occupied by this convective core boundary as a function of the star's initial mass at TAMS.

i) Behavior as a Function of Mass

The graph displays a distinct, monotonic upward trend where the upper boundary of the convective core (q_{top}) scales directly with increasing initial stellar mass:

- **Lower High-Mass Stars ($\sim 10 M_{\odot}$):** The convective core at TAMS is relatively compressed, extending to only $q \approx 0.13$, meaning the core contains the innermost 13% of the star's mass.

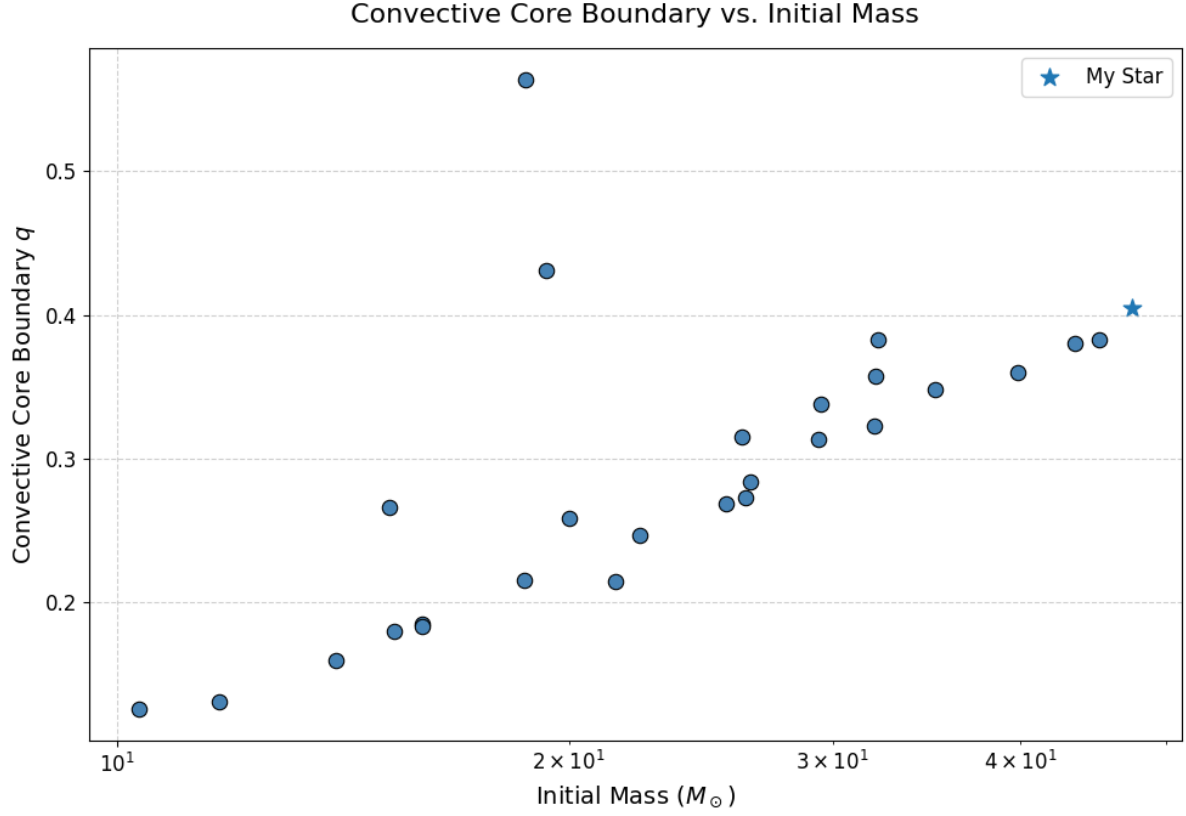


Figure 9: The convective core boundary coordinate (q_{top}) at the Terminal-Age Main Sequence (TAMS) plotted against the initial stellar mass ($M/M_{\odot}$) for the compiled class dataset. Because every star in this high-mass sample possesses a convective core originating at the center ($q_{\text{bottom}} = 0$), the y-axis directly measures the total mass fraction occupied by the convective core at the moment of core hydrogen exhaustion.

- **Intermediate High-Mass Stars ($\sim 20 - 30 M_{\odot}$):** The core boundary moves steadily outward, with the convective core encompassing roughly 25% to 30% of the total stellar mass ($q \approx 0.25 - 0.30$).
- **Extreme High-Mass Stars ($\sim 45 - 50 M_{\odot}$):** The convective core boundary reaches its maximum extent, capping out between $q \approx 0.38$ and 0.40 . The $47.504 M_{\odot}$ model analyzed in this work aligns perfectly with this class distribution, terminating its main sequence with a convective core at $q = 0.405$.

This behavior is dictated by the microphysics of the core energy generation source. Massive stars are powered almost entirely by the Carbon-Nitrogen-Oxygen (CNO) cycle. The nuclear energy generation rate of the CNO cycle depends exceptionally strongly on temperature, scaling approximately as:

$$\epsilon_{\text{CNO}} \propto T^{15-20} \quad (12)$$

As initial stellar mass increases, the central temperature climbs. Due to this extreme power-law dependence, the nuclear energy generation becomes highly concentrated toward the absolute center of the star. This massive energy release within a small central volume generates a colossal local radiative flux that cannot be transported by radiation alone. The resulting sharp increase in the radiative temperature gradient (∇_{rad}) causes a larger overall mass fraction of the star's inner core to satisfy the Schwarzschild criterion ($\nabla_{\text{rad}} > \nabla_{\text{ad}}$), pushing the convective boundary (q_{top}) farther outward.

ii) Confrontation with Figure 8.8 (ZAMS vs. TAMS Structures)

When comparing the compiled class dataset at TAMS to the classic diagram for the Zero-Age Main Sequence (ZAMS) shown in Figure 8.8 of the SSE lecture notes, a major structural difference is apparent: **the convective cores in these TAMS models are systematically smaller in mass fraction (q) than those expected at ZAMS for identical initial masses.**

For instance, Figure 8.8 demonstrates that a star near $45 M_{\odot}$ begins its life on the ZAMS with an expansive convective core enclosing well over 60% of its total mass ($q \approx 0.60$). However, our class dataset shows that by the time these stars reach TAMS, their cores have shrunk down to $q \approx 0.38 - 0.40$.

The smaller convective-core mass fractions observed across the class dataset therefore provide clear empirical evidence of the progressive recession of the convective core during core hydrogen burning.

2.3 HR Diagram Comparison (Q2.3)

The Hertzsprung-Russell (HR) diagram shown in Figure 10 demonstrates the fundamentally divergent evolutionary pathways followed by high-mass and low-mass stars after leaving the main sequence.

The $47.504 M_{\odot}$ model commences its main sequence evolution as an exceptionally hot, highly luminous O-type star ($\log T_{\text{eff}} \approx 4.65$, $\log L/L_{\odot} \approx 5.5$). Upon exhausting its central hydrogen supply at the TAMS, the star experiences a negligible shift in total luminosity, evolving nearly horizontally across the HR diagram toward lower effective temperatures ($\log T_{\text{eff}} \approx 3.56$). This horizontal migration reflects a rapid, non-equilibrium expansion of the stellar envelope into a cold, red supergiant configuration, driven by the contraction of its non-degenerate helium core.

Conversely, the low-mass $0.665 M_{\odot}$ model evolves on a dramatically different timescale and geometric path. Following core hydrogen exhaustion, the star undergoes structural reorganization, causing its outer layers to expand and cool. It then experiences a steep vertical climb up the red giant branch (RGB), where its effective temperature stabilizes around $\log T_{\text{eff}} \approx 3.65$ while its luminosity surges upward by orders of magnitude to $\log L/L_{\odot} \approx 1.82$.

The variation in $\Delta \log L$ post-main sequence is caused by the different degrees of electron degeneracy within their respective cores. Massive stars maintain a hot, non-degenerate helium core that contracts

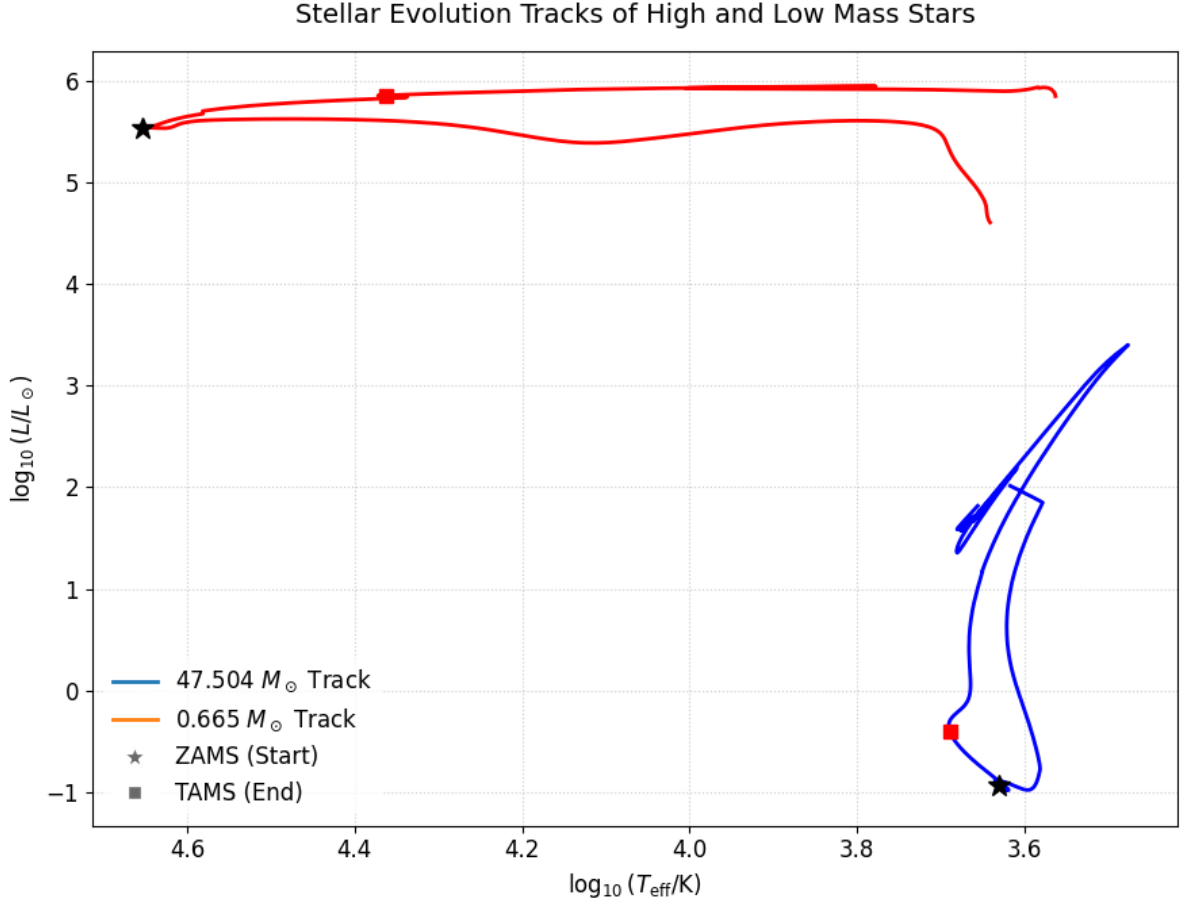


Figure 10: Comparative Hertzsprung-Russell (HR) diagram plotting the evolutionary tracks ($\log L/L_{\odot}$ vs. $\log T_{\text{eff}}/\text{K}$) for the high-mass $47.504 M_{\odot}$ model (red path) and the low-mass $0.665 M_{\odot}$ model (blue path). Chronological key points are explicitly marked, including initial Zero-Age Main Sequence conditions (ZAMS, filled stars) and final Terminal-Age Main Sequence conditions (TAMS, filled squares). The high-mass model evolves horizontally toward cold temperatures at an almost constant, extreme luminosity, whereas the low-mass model exhibits a steep, vertical ascent up the red giant branch.

smoothly without triggering degeneracy, allowing the star to ignite shell burning and advanced core helium burning at a nearly stable total radiative output. In contrast, low-mass stars develop highly dense, electron-degenerate helium cores. The degenerate core cannot expand to cool itself, causing the surrounding hydrogen-burning shell to intensify dramatically. This shell burning injects energy into the non-degenerate outer envelope, forcing an expansion that drives the major vertical luminosity climb observed along the red giant branch.

2.4 Central Temperature–Density Topology (Q2.4)

Tracking stellar core conditions within a central temperature vs. central density ($\log T_c$ vs. $\log \rho_c$) diagnostic framework isolates the governing equation of state (EOS) driving stellar lifecycles. The standard thermodynamic boundaries separating distinct pressure regimes are derived analytically by equating competing pressure terms.

The transition boundary separating ideal gas pressure domination from radiation pressure domination is defined by setting:

$$P_{\text{gas}} = P_{\text{rad}} \implies \frac{\rho k_B T}{\mu m_u} = \frac{1}{3} a T^4 \implies T \propto \rho^{1/3} \quad (13)$$

On a log-log plot, this boundary maps as a straight line with a characteristic slope of $1/3$.

Similarly, the boundary separating non-degenerate ideal gas behavior from non-relativistic electron degeneracy pressure is established via:

$$P_{\text{gas}} = P_{\text{deg,NR}} \implies \frac{\rho k_B T}{\mu m_u} = K_1 \left(\frac{\rho}{\mu_e} \right)^{5/3} \implies T \propto \rho^{2/3} \quad (14)$$

This renders a linear boundary possessing a steeper slope of $2/3$. Finally, the extreme transition threshold into relativistic electron degeneracy occurs when degenerate Fermi momentum scales near relativistic limits, taking place at a critical, nearly constant density threshold ($\rho_{\text{crit}} \sim 10^6 \text{ g cm}^{-3}$), which visually represents a vertical boundary line.

Figure 11 underscores the contrasting core physical conditions. The $47.504 M_\odot$ model tracks through low-density, high-temperature regimes. Its core avoids the electron degeneracy zone entirely, moving safely within ideal gas regimes while trending closer toward radiation pressure domination ($\log T_c = 8.48$, $\log \rho_c = 3.05$ at final helium depletion).

In contrast, the $0.665 M_\odot$ model tracks to high densities before reaching advanced ignition temperatures. Its path crosses deep into the non-relativistic electron degeneracy regime, emphasizing that degeneracy pressure provides the primary structural support against gravitational collapse throughout its post-main sequence life, up until helium ignition ($\log T_c = 8.20$, $\log \rho_c = 4.63$).

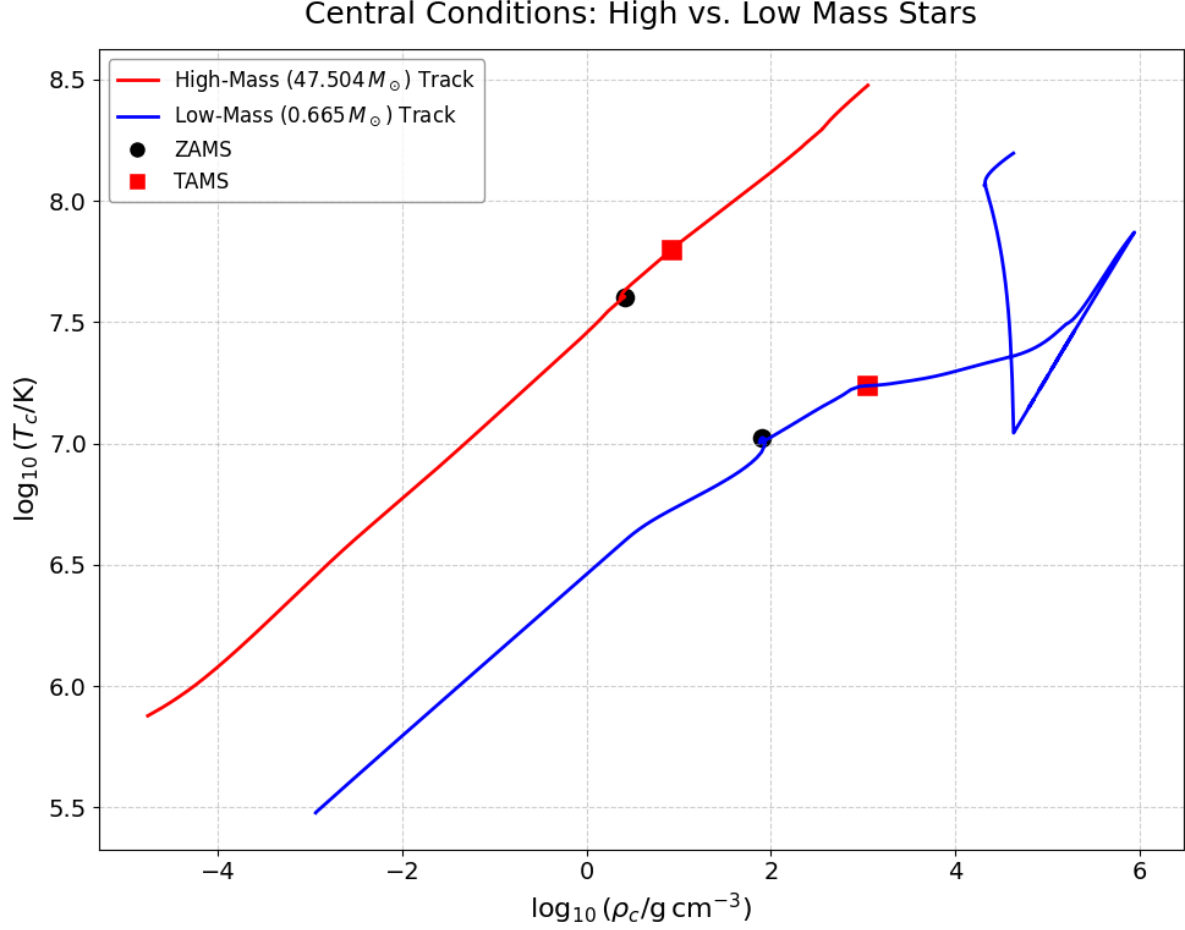


Figure 11: Stellar core evolution pathways tracked within the central temperature vs. central density ($\log_{10} T_c/\text{K}$ vs. $\log_{10} \rho_c/\text{g cm}^{-3}$) plane. The evolutionary trajectories of the high-mass $47.504 M_\odot$ model (red line) and the low-mass $0.665 M_\odot$ model (blue line) are compared directly alongside their respective ZAMS (black circle) and TAMS (red square) coordinates. The low-mass star shifts rightward into high-density, degenerate regimes, while the high-mass track stays within ideal gas and radiation-dominated territory.

2.5 Summary Data Appendix

The precise global and core physical parameters extracted from the final MESA simulation profiles are compiled systematically in Table 2.

Table 2: Comprehensive Physical Parameters Extracted from MESA Simulation Phases

Physical Parameter	High-Mass (1 st Phase)	High-Mass (2 nd Phase)	Low-Mass Model
Evolutionary Criterion	Core H ₁ Exhaustion	Central He ₄ → 0.01	Central He ₄ → 0.01
Final Model Number	1055	2799	21706
Stellar Age [yrs]	4.07×10^6	4.46×10^6	5.48×10^{10}
Central H ₁ Fraction	0.0087	0.0000	0.0000
Central He ₄ Fraction	0.9719	0.0100	0.0097
$\log(L/L_{\odot})$	5.8350	5.8436	1.8156
$\log(T_{\text{eff}}/\text{K})$	4.3372	3.5631	3.6559
$\log(T_c/\text{K})$	7.7294	8.4764	8.1963
$\log(\rho_c/\text{g cm}^{-3})$	0.7093	3.0521	4.6313

3 Theoretical Framework and Numerical Implementation

3.1 Convection Criteria and Stability (Q3.1)

To evaluate local hydrodynamic stability against convection within the non-homogeneous layers of a stellar interior, standard stellar structure theory relies on two primary analytical formulations: the Schwarzschild criterion and the Ledoux criterion.

The simpler formulation, the **Schwarzschild criterion**, considers purely thermal buoyancy forces within a chemically uniform layer. An interior cell becomes dynamically unstable to turbulent convective mixing whenever the local radiative temperature gradient (∇_{rad}) exceeds the adiabatic temperature gradient (∇_{ad}):

$$\nabla_{\text{rad}} > \nabla_{\text{ad}} \quad (15)$$

In regions where nuclear burning or microscopic settling establishes a spatial composition gradient, the changing mean molecular weight acts as a stabilizing or destabilizing agent. To account for this chemical stratification, the **Ledoux criterion** introduces a composition gradient term, defining instability via:

$$\nabla_{\text{rad}} > \nabla_{\text{ad}} + \frac{\phi}{\delta} \nabla_{\mu} \quad (16)$$

where $\nabla_{\mu} = \frac{d \ln \mu}{d \ln P}$ represents the structural mean molecular weight gradient, while ϕ and δ correspond to the standard thermodynamic partial derivatives evaluated from the equation of state:

$$\phi = \left(\frac{\partial \ln \rho}{\partial \ln \mu} \right)_{P,T}, \quad \delta = - \left(\frac{\partial \ln \rho}{\partial \ln T} \right)_{P,\mu} \quad (17)$$

When a star develops a heavy, helium-rich core beneath a lighter hydrogen envelope, a positive ∇_{μ} gradient is established ($\nabla_{\mu} > 0$). This acts as an extra stabilizing barrier, requiring a significantly higher radiative gradient to trigger convection compared to the pure Schwarzschild limit.

In these simulations, the MESA (Modules for Experiments in Stellar Astrophysics) software was left at its default configuration, which utilizes the **Schwarzschild criterion** as the definitive boundary selector for convective zone allocation throughout both main-sequence and core-helium burning evolutionary phases.

3.2 The Reality of Convective Overshooting (Q3.2)

In classical stellar theory, we define the edge of a convective core strictly by the Schwarzschild criterion ($\nabla_{\text{rad}} = \nabla_{\text{ad}}$)—the exact boundary where the net buoyancy force on a rising fluid element drops to zero. Real life, however, is rarely so tidy. Turbulent fluid elements churning inside a massive core carry significant momentum. When they reach this formal boundary, their inertia carries them past it, driving them deep into the stable, sub-adiabatic radiative layers directly above before buoyancy can slow them down and reverse their direction. We call this phenomenon convective overshooting.

In our MESA models, this non-local, multi-dimensional turbulent penetration is parameterized as an exponential diffusion process. The mixing coefficient (D_{ov}) in the overshooting region drops off as a function of distance (z) from the Schwarzschild edge according to:

$$D_{\text{ov}}(z) = D_0 \exp\left(-\frac{2z}{f_{\text{ov}}H_P}\right) \quad (18)$$

where D_0 is the diffusion coefficient evaluated right at the convective boundary, H_P is the local pressure scale height, and f_{ov} is a free, dimensionless scaling parameter that regulates how deep the extra chemical mixing penetrates.

Injecting this extra overshooting mixing fundamentally changes how a massive star lives and grows:

1. **Expanding the Core:** It pulls fresh hydrogen fuel from the radiative envelope down into the central burning engine, meaning the star maintains a significantly larger mixed convective core mass fraction (M_{cc}/M_*) throughout its main-sequence life.
2. **Extending Lifetimes:** A larger reservoir of core hydrogen means the star has a much bigger fuel supply, which significantly stretches out its main-sequence lifetime (τ_{MS}).
3. **Altering Global Properties:** Because more fuel is consumed during the main sequence, the star leaves behind a much heavier helium core (M_{He}). This extra core mass drives up the stellar luminosity (L) and causes the main sequence to broaden significantly, allowing the star to expand to cooler effective temperatures (T_{eff}) and lower surface gravities ($\log g$) before core hydrogen is exhausted.

To study these effects, we ran a grid of $12 M_{\odot}$ rotating stellar models ($v_{\text{rot}} = 400 \text{ km s}^{-1}$) from the ZAMS down to a remaining core hydrogen fraction of $X_c = 0.04$. We varied the overshooting parameter across four discrete values: $f_{\text{ov}} = 0.000, 0.050, 0.100$, and 0.258 . The final properties of these models are highlighted in Table 3.

Table 3: Properties of our $12 M_{\odot}$ model grid evaluated at $X_c = 0.04$.

f_{ov}	Convective Core Fraction	Stellar Age [Myr]	$\log(L/L_{\odot})$	$\log(T_{\text{eff}}/\text{K})$	$\log(g/\text{cm s}^{-2})$
0.000	0.154	14.76	4.263	4.329	3.524
0.050	0.240	20.33	4.523	4.244	2.925
0.100	0.340	23.66	4.737	4.136	2.277
0.258	0.564	28.33	5.047	4.035	1.565

The trends in the table are striking. Turning up the overshooting efficiency dramatically increases the convective core mass fraction, nearly doubling the star’s main-sequence lifespan from 14.76 Myr to 28.33 Myr. At the same time, the final luminosity climbs significantly while the surface gravity drops. In fact, our largest overshooting model ($f_{\text{ov}} = 0.258$) maintains a convective core that is more than three

times larger than the model with no overshooting at all, allowing the star to expand until its surface gravity falls to $\log g = 1.565$.

To visualize this dramatic atmospheric divergence, we can look at the Kiel diagram ($\log T_{\text{eff}} - \log g$) in Figure 12. Here, we directly contrast the standard track calculated without overshooting ($f_{\text{ov}} = 0$) against our heavily mixed model ($f_{\text{ov}} = 0.258$).

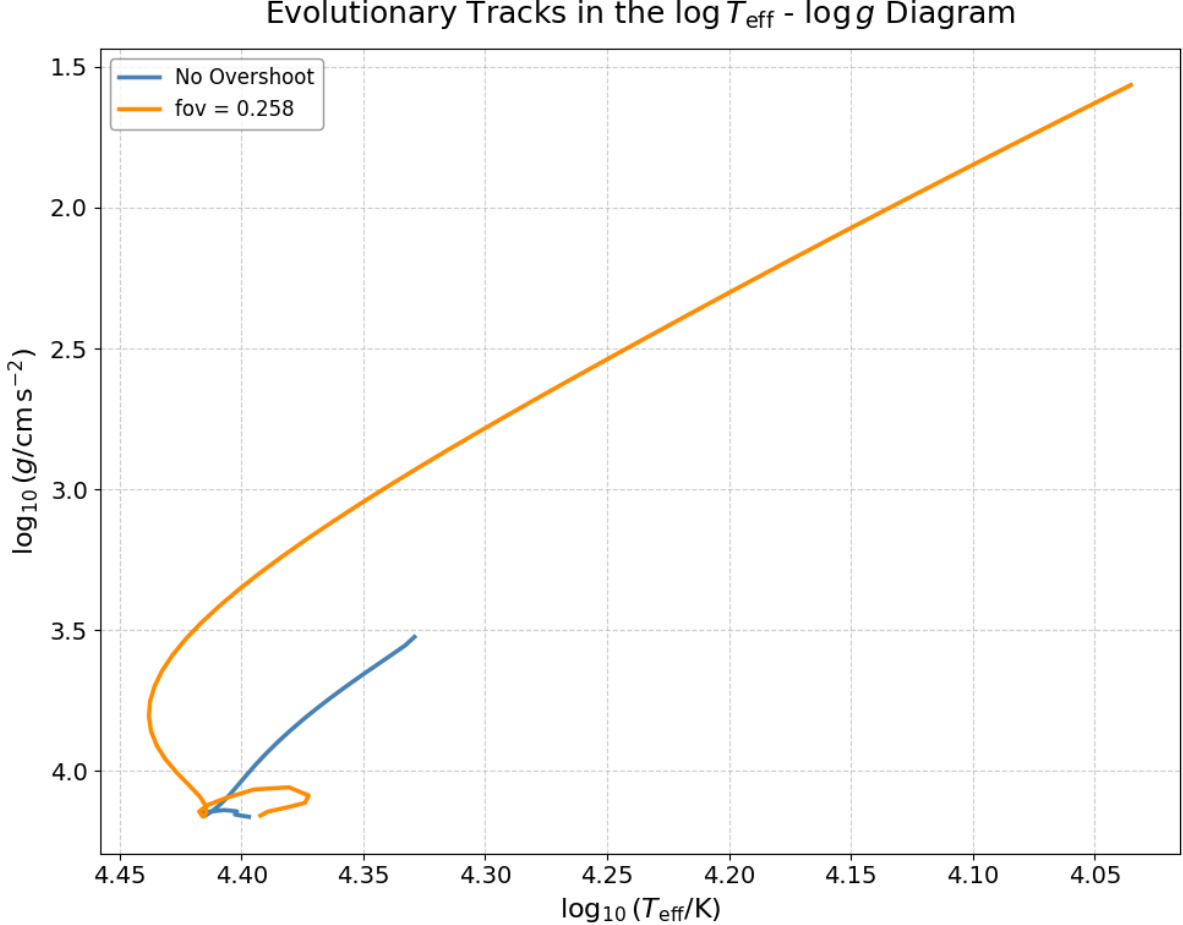


Figure 12: Kiel diagram ($\log_{10}(g/\text{cm s}^{-2})$ vs. $\log_{10}(T_{\text{eff}}/\text{K})$) comparing the evolutionary pathways of our $12 M_{\odot}$ grid. The solid blue line marks the track computed without extra boundary mixing ($f_{\text{ov}} = 0$), which terminates at a high surface gravity. The red curve tracks the extreme overshooting model ($f_{\text{ov}} = 0.258$), showcasing how a massive core allows the star to expand into much lower surface gravities and colder effective temperatures.

3.3 Calibrating Overshooting Against Observations (Q3.3)

Because we cannot look directly inside a star to watch turbulent fluid elements overshoot, we have to constrain f_{ov} indirectly by matching our evolutionary models to what we see in the night sky.

Our calibration method relies on plotting projected surface rotation speeds ($v \sin i$) against surface gravities ($\log g$) for a large sample of massive stars. Massive stars are born as rapid rotators on the ZAMS. As they move across the main sequence, core contraction combined with envelope expansion causes their overall radius to swell. By conservation of angular momentum, this structural inflation forces their surface rotation rates to drop. Once core hydrogen is completely exhausted, the star leaves the main sequence and expands rapidly, causing its surface rotation to drop off a cliff. The sudden

disappearance of rapidly rotating stars in an observational sample therefore marks the structural end of the main sequence (the TAMS).

To find this empirical boundary, we isolated stars with clear, rapid rotation ($v \sin i > 80 \text{ km s}^{-1}$) from the dataset. Figure 13 displays a histogram of these fast-rotating stars distributed across surface gravity bins.

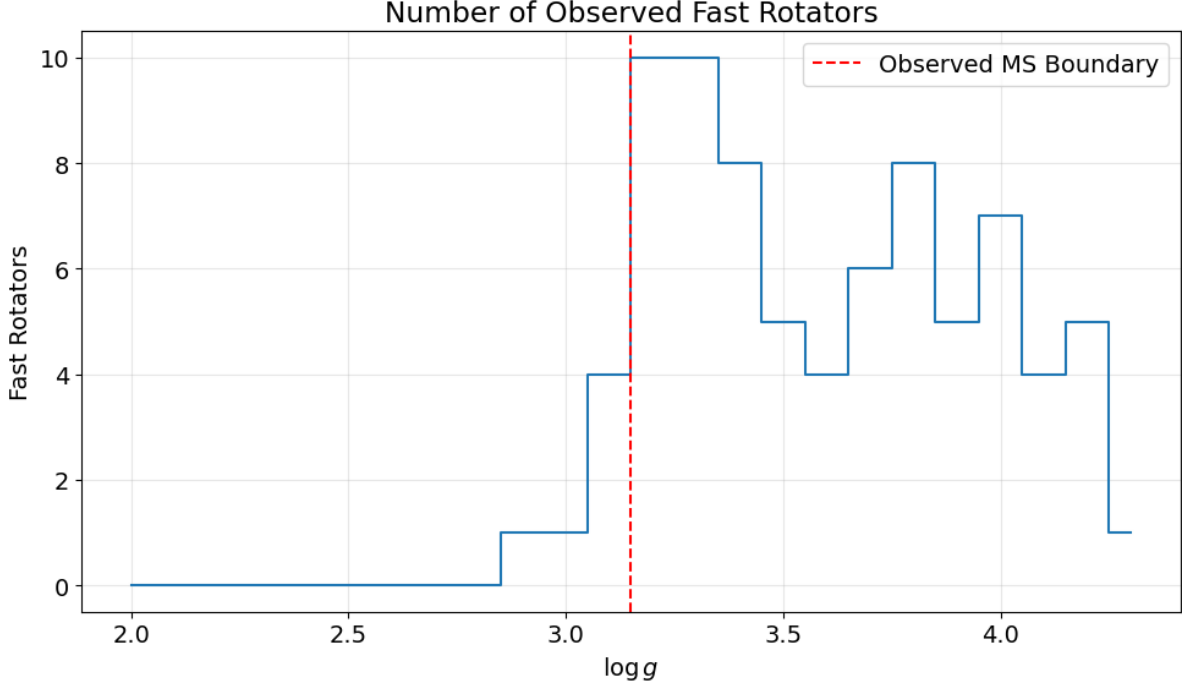


Figure 13: Distribution of rapidly rotating stars ($v \sin i > 80 \text{ km s}^{-1}$) as a function of surface gravity ($\log g$). The sharp decline in star counts below $\log g \approx 3.15$ marks the empirical edge of the terminal-age main sequence, where rapid rotation ceases due to major post-main-sequence radius expansion.

The distribution shows a sharp, unambiguous drop-off in the number of fast rotators at lower surface gravities, centered right around:

$$\log g_{\text{obs,TAMS}} \approx 3.15 \quad (19)$$

We adopt this value as our observational TAMS proxy.

To calibrate our overshooting parameter, we compare this empirical value ($\log g \approx 3.15$) to the surface gravities reached by our MESA models at core hydrogen exhaustion ($X_c = 0.04$):

$$\log g_{\text{TAMS}}(f_{\text{ov}}) = \begin{cases} 3.524 & \text{for } f_{\text{ov}} = 0.000 \\ 2.925 & \text{for } f_{\text{ov}} = 0.050 \\ 2.277 & \text{for } f_{\text{ov}} = 0.100 \\ 1.565 & \text{for } f_{\text{ov}} = 0.258 \end{cases} \quad (20)$$

As we expect, increasing f_{ov} widens the main sequence and pushes the final TAMS surface gravity systematically lower. The observed boundary value of 3.15 sits directly between our non-overshooting model (3.524) and our moderate overshooting model (2.925).

To pin down the exact value of f_{ov} that best reproduces the data, we perform a linear interpolation between the $f_{\text{ov}} = 0.000$ and $f_{\text{ov}} = 0.050$ models:

$$f_{\text{ov}} = 0.050 \times \frac{3.524 - 3.15}{3.524 - 2.925} = 0.050 \times \frac{0.374}{0.599} \approx 0.031 \quad (21)$$

This interpolation shows that an overshooting parameter of:

$$f_{\text{ov}} \approx 0.03 \quad (22)$$

provides the best fit to the observed width of the main sequence. Figure 14 maps this relationship directly, plotting our modeled TAMS surface gravities across our continuous parameter grid alongside the empirical cutoff line.

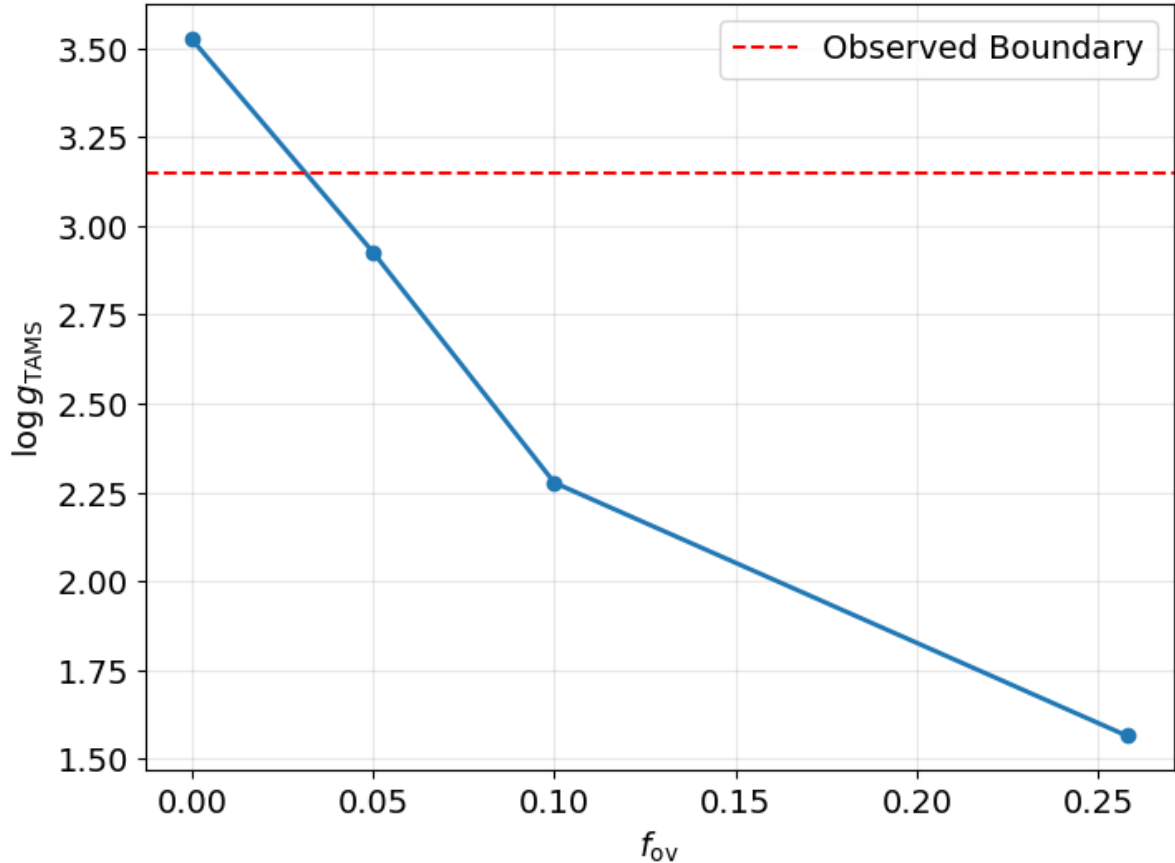


Figure 14: Dependence of the terminal-age main-sequence surface gravity ($\log g_{\text{TAMS}}$) on the convective overshooting parameter f_{ov} . The horizontal dashed line marks the empirical main-sequence boundary ($\log g = 3.15$). The intersection pinpointing where our models match the data yields a best-fit overshooting parameter of $f_{\text{ov}} \approx 0.031$.

Summary of Convective Overshooting Calibration

Convective overshooting exerts a massive, unavoidable control on how massive stars live and evolve by physically altering the size of their mixed core. Larger overshooting values yield longer main-sequence lifetimes, heavier helium cores, higher overall luminosities, and much greater radius expansion.

Comparing our evolutionary models with the observed surface gravities of rapidly rotating stars demonstrates that the empirical width of the main sequence is beautifully reproduced by an overshooting parameter of approximately $f_{\text{ov}} \approx 0.03$. Models with larger overshooting parameters ($f_{\text{ov}} \geq 0.10$, and especially the extreme case of $f_{\text{ov}} = 0.258$) keep the core stable against exhaustion for far too long. They push the main sequence deep into low surface gravity territory, creating wide tracks that are completely contradicted by the rotation cutoffs we observe in nature.

4 Nuclear Networks and Pre-Main-Sequence Deuterium Fusion

4.1 Isotopic Extensions to the Proton-Proton Chain (Q4.1)

To investigate the structural and energetic role of light elements during early stellar life, our baseline model of a $0.665 M_{\odot}$ star was switched from a basic nuclear network to the more comprehensive `pp_extras.net`.

Inspecting the configuration file reveals that the standard proton-proton (pp) chain is augmented by explicitly adding four light isotopes:

$$\text{add_isos}(\text{h2}, \text{li7}, \text{be7}, \text{b8}) \quad (23)$$

These additions represent Deuterium (^2H), Lithium-7 (^7Li), Beryllium-7 (^7Be), and Boron-8 (^8B). While all of these elements participate as intermediate steps within the branches of the pp-chain, tracking Deuterium (^2H) is particularly useful here. It allows us to explicitly monitor the initial nuclear burning phase that occurs long before the main sequence begins.

4.2 Pre-Main-Sequence Contraction and the Deuterium Thermostat (Q4.2)

A star begins its life as a large, cold, diffuse cloud that must contract under gravity to heat up its core. Figure 15 maps this contraction path, showing how the stellar radius evolves against time during the pre-main-sequence (PMS) phase.

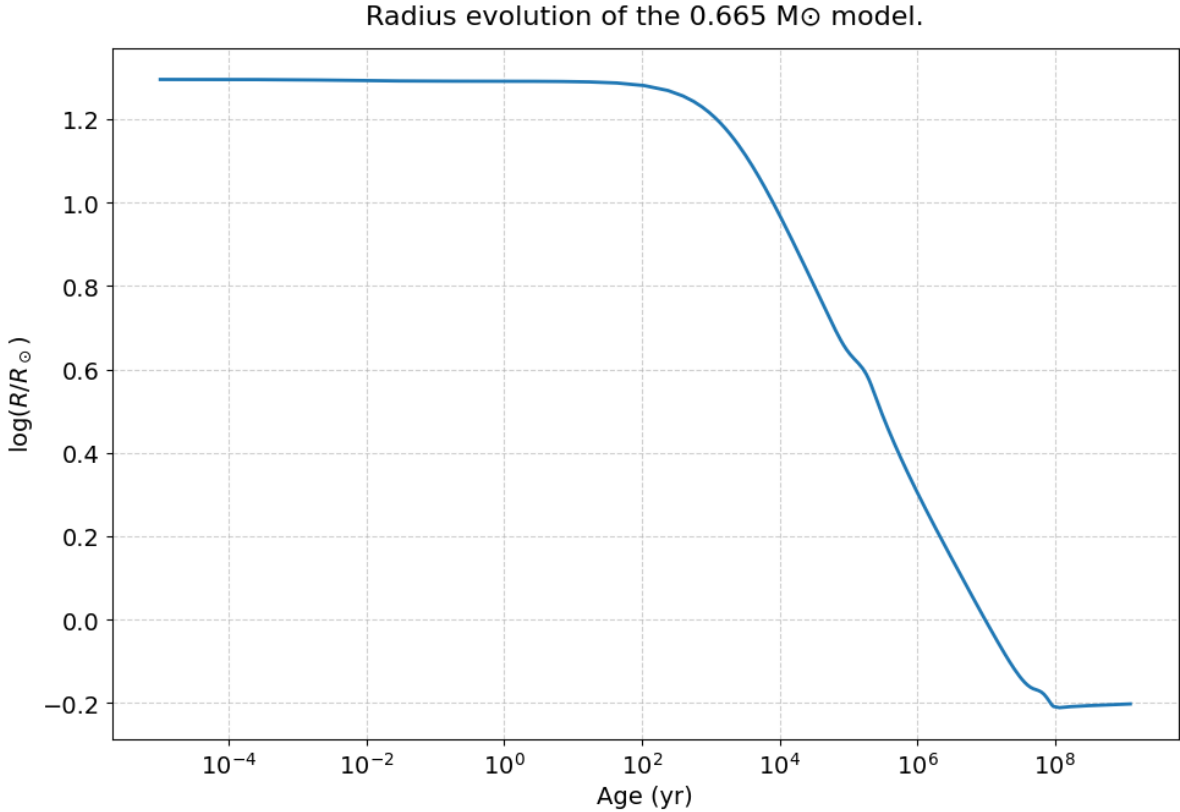


Figure 15: Pre-main-sequence radius evolution ($\log_{10}(R/R_{\odot})$ vs. Age) for the $0.665 M_{\odot}$ stellar model. The star undergoes structural contraction, pausing briefly during early phases due to the energetic support of core deuterium burning before continuing down to its stable main-sequence configuration.

The simulation captures a dramatic structural collapse. The model starts out as a bloated protostar with an initial radius of:

$$\log(R/R_{\odot}) = 1.295 \implies R_{\text{initial}} \approx 19.7 R_{\odot} \quad (24)$$

By the time contraction slows near the ZAMS, the radius has shrunk by more than an order of magnitude to:

$$\log(R/R_{\odot}) = -0.202 \implies R_{\text{final}} \approx 0.63 R_{\odot} \quad (25)$$

As the star compresses, gravitational potential energy is converted directly into heat. This drives the central temperature (T_c) upward from its icy baseline of $\log T_c = 5.48$ ($T_c \approx 3 \times 10^5$ K) to a blistering $\log T_c = 7.00$ ($T_c \approx 10^7$ K), at which point standard hydrogen fusion can permanently take over.

However, unlike the simplified model we ran in Exercise 2, this model includes explicit deuterium burning via the proton-capture reaction:



Because this reaction ignites at much lower temperatures than standard hydrogen fusion, it provides an unexpected burst of core nuclear energy right in the middle of the pre-main-sequence phase. This localized energy release acts as a structural thermostat. The outward nuclear pressure partially counteracts the gravitational collapse, temporarily slowing down the star's contraction rate. The distinct flattening or change in slope seen in the radius track (Figure 15) is a direct visual signature of primordial deuterium temporarily holding gravity at bay.

4.3 The Spatial Dynamics of Deuterium Depletion (Q4.3)

A key question in early stellar evolution is whether this primordial fuel burns away uniformly throughout the star or vanishes in a sudden flash. To track this, we monitored a sequence of internal structural profiles (Profiles 60 through 80) around the exact moment of depletion. Figure 16 illustrates the spatial abundance of deuterium (X_2) across the mass coordinate of the star during this transition.

Looking at the peak values across the profiles reveals an incredibly rapid drop-off:

$$X({}^2\text{H})_{\text{max}} = \begin{cases} 1.0 \times 10^{-5} & \text{for Profiles 60–65} \\ 1.75 \times 10^{-17} & \text{for Profile 66} \\ 1.79 \times 10^{-17} & \text{for Profile 80} \end{cases} \quad (27)$$

Between Profile 65 and Profile 66, the global deuterium abundance plummets by a staggering 12 orders of magnitude, effectively vanishing from the output profiles in a single step.

While the data looks like an instantaneous explosion, this abrupt drop is largely a side effect of finite temporal resolution in our saved profile outputs. In reality, stellar physics dictates that deuterium burns progressively. The reaction ignites first in the absolute center where temperatures are highest, and then eats its way outward to larger mass coordinates as the rest of the interior warms up.

The reason it appears to disappear everywhere at once is due to the structure of a $0.665 M_{\odot}$ star at this age. The star is completely convective during this phase, functioning like a giant, turbulent blender. As the core reaches the ignition threshold of $\approx 10^6$ K and rapidly destroys its local deuterium, convection instantly churns fresh, primordial material down from the outer layers into the central furnace. This highly efficient mixing cycle completely drains the star's entire deuterium reservoir over a very short evolutionary window, which registers as a sudden, uniform disappearance in our simulation timestamps.

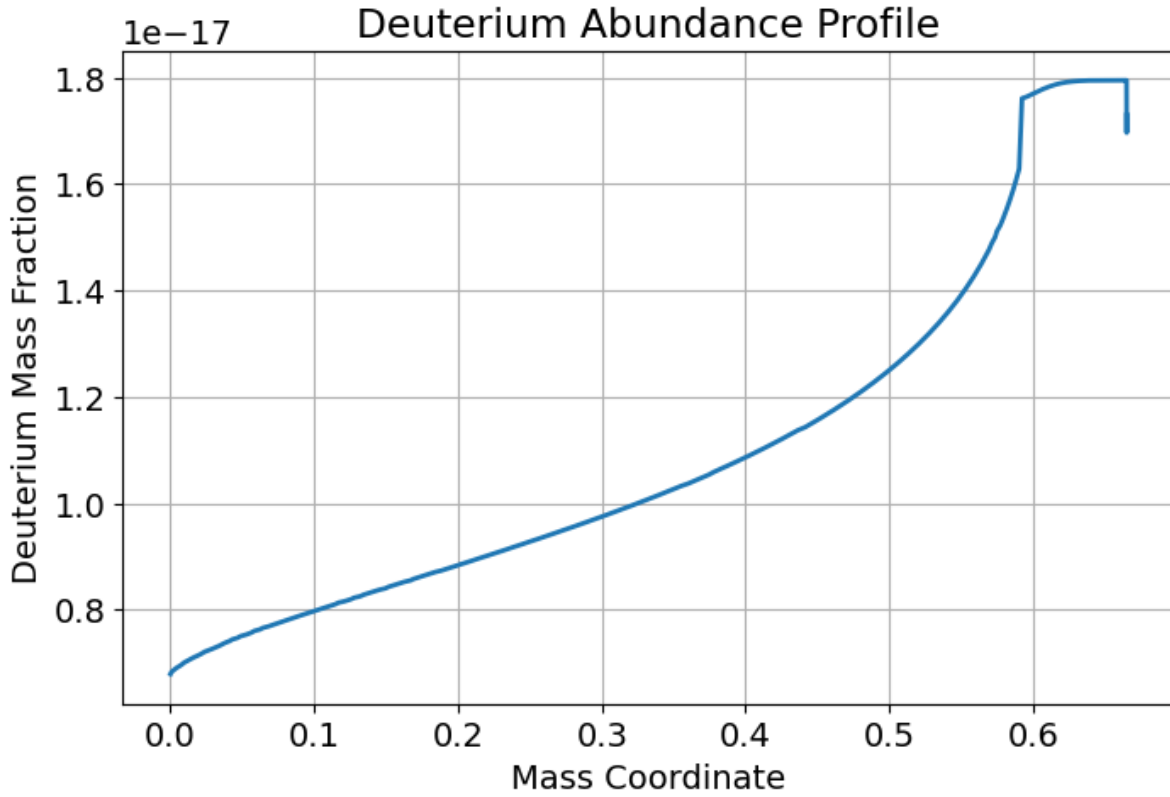


Figure 16: Deuterium mass fraction abundance profiles (X_D) plotted against the internal mass coordinate ($M/M_\odot$) across the depletion phase. The primordial abundance remains stable through Profile 65 before plunging cleanly below 10^{-16} by Profile 66 due to ignition and convective mixing.

4.4 Stellar Evolution in an Alternative Universe: Hot Deuterium Burning (Q4.4)

To explore the sensitivity of this early evolutionary phase, we consider an alternative universe where a tweak to nuclear physics raises the required deuterium ignition temperature by nearly an order of magnitude, scaling it up by a factor of $k = 8.5$:

$$T_{\text{burn}} = 8.5 \times 10^6 \text{ K} \quad (\text{instead of } 10^6 \text{ K}) \quad (28)$$

To understand how this changes the star's internal chemistry, we can examine the final internal temperature profile generated by our code, shown in Figure 17. Horizontal reference lines are overlaid at both the standard threshold (10^6 K) and the hypothetical alternative threshold ($8.5 \times 10^6 \text{ K}$).

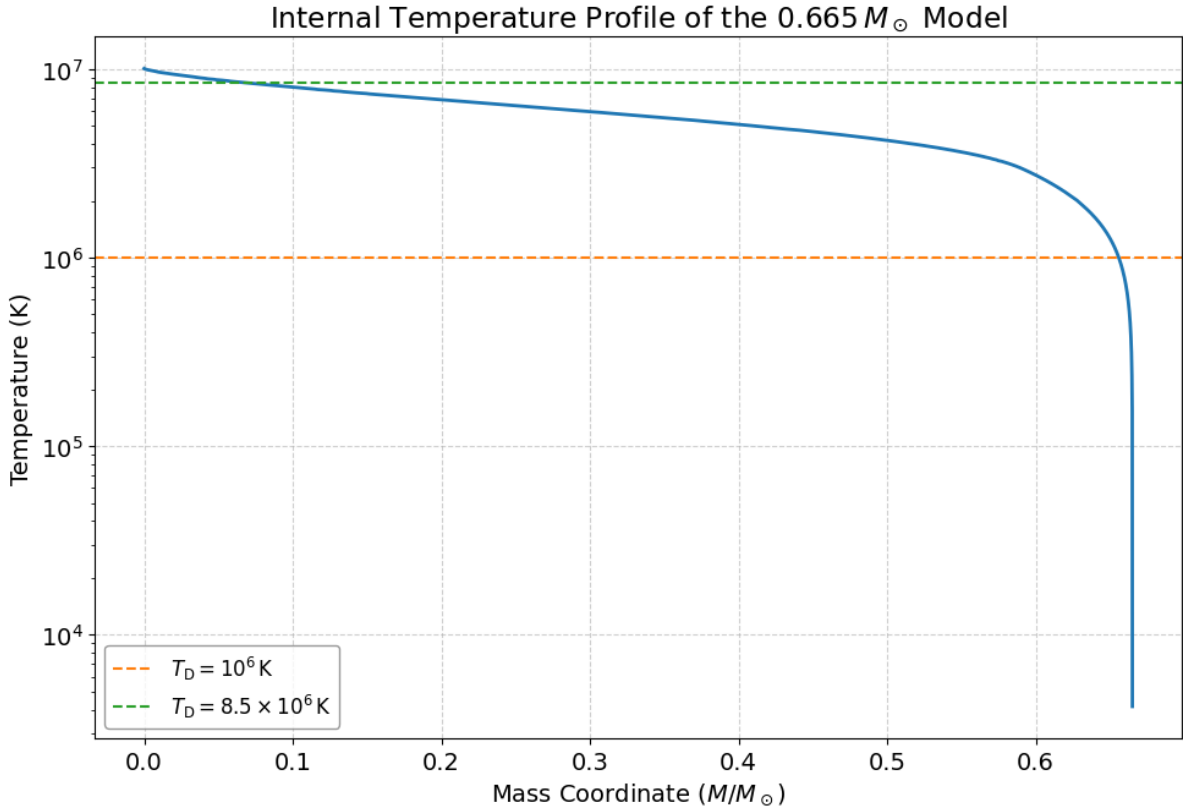


Figure 17: Internal temperature profile (T in Kelvin) vs. mass coordinate ($M/M_{\odot}$) for the $0.665 M_{\odot}$ model. The horizontal dashed lines mark the standard ignition threshold (10^6 K) and our alternative universe threshold ($8.5 \times 10^6 \text{ K}$), illustrating that only a fraction of the deep core ever reaches the hotter burning limit.

The core of our star eventually tops out at $\log T_c \approx 7.00$ ($T_c \approx 10^7 \text{ K}$). Looking at the temperature profile in Figure 17, a stark spatial contrast becomes clear:

- **Standard Universe (10^6 K):** Virtually the entire stellar interior easily exceeds this temperature. Consequently, any deuterium mixed into almost any part of the interior is cleanly incinerated.
- **Alternative Universe ($8.5 \times 10^6 \text{ K}$):** Only the innermost central core ($M/M_{\odot} < 0.15$) ever manages to climb past this high thermal barrier. The middle and outer layers of the star remain permanently below the required ignition threshold.

This shift would alter the chemical landscape of the star. Instead of a rapid, global cleanup of deuterium, burning would be confined strictly to a small, localized zone at the absolute center. Because the outer layers never get hot enough to ignite the reaction, they would preserve their primordial deuterium abundance completely intact.

Furthermore, as the star continues to contract and age, the core shrinks and convection eventually retreats, locking this unburned fuel away in the envelope. Rather than the uniform, complete depletion we observe in our universe, the alternative universe would produce a star with a steep, permanent radial abundance gradient, leaving a wealth of unburned primordial deuterium stranded in its outer layers.

4.5 Global Abundance Summary

To see how these nuclear interactions reshape the star's composition over time, the final chemical state of the model is illustrated in Figure 18. This appendix tracks the surviving mass fractions of Hydrogen, Deuterium, Helium-3, and Helium-4 across the entire stellar mass coordinate at the conclusion of our run.

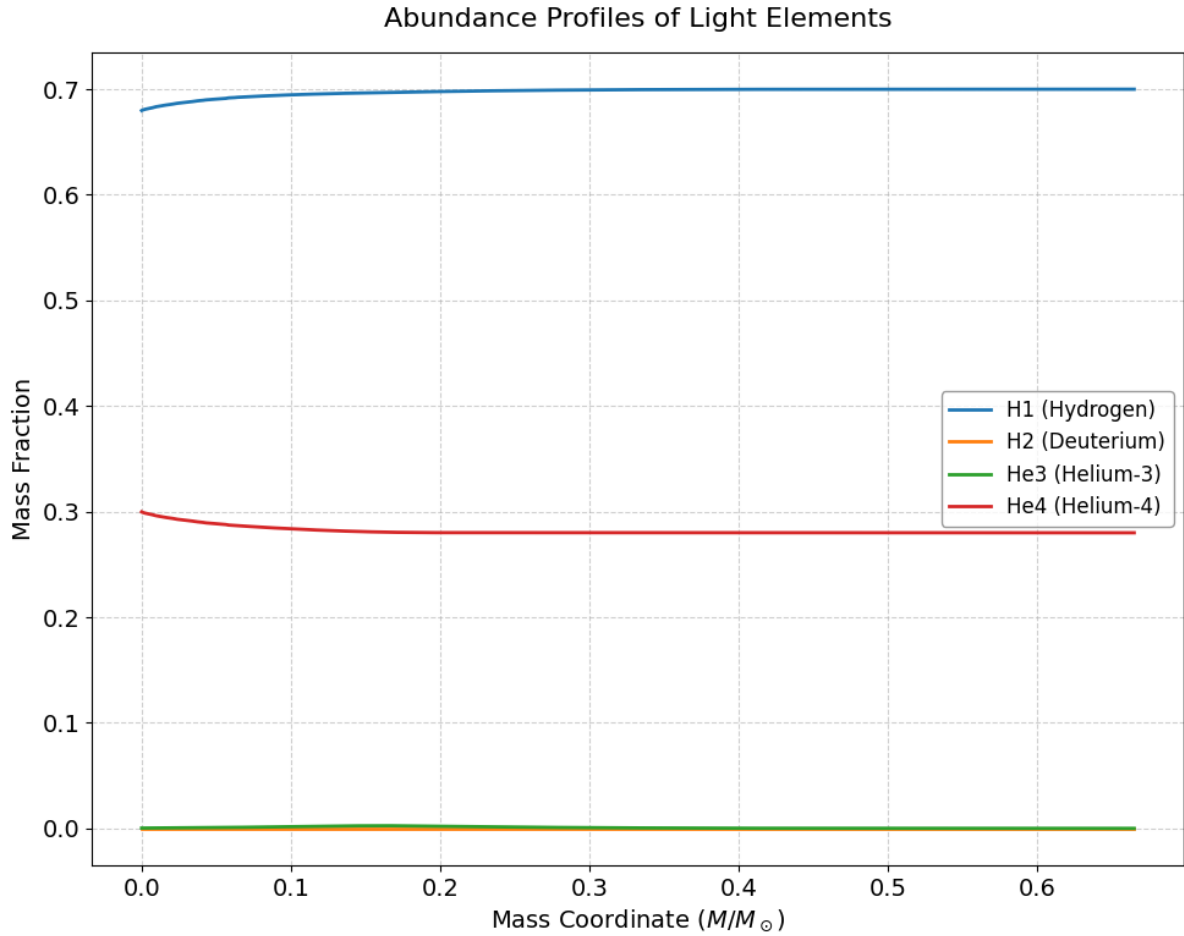


Figure 18: Final light-element abundance profiles (Mass Fraction vs. Mass Coordinate $M/M_{\odot}$) for the $0.665 M_{\odot}$ star. The completely flat, near-zero baseline of Deuterium (^2H) confirms total global destruction across the convective layers, leaving behind a cleanly mixed, hydrogen-rich envelope.